Suggested Answers to Problem Set 3

1. If β is not restricted, under $H_0 : \delta = 0$ we have

$$\begin{aligned} y_t &= \alpha + \beta t + y_{t-1} + \varepsilon_t \\ &= \alpha + \beta t + [\alpha + \beta (t-1) + y_{t-2} + \varepsilon_{t-1}] + \varepsilon_t \\ &= 2\alpha + \beta [t + (t-1)] + y_{t-2} + \varepsilon_{t-1} + \varepsilon_t \\ &= 3\alpha + \beta [t + (t-1) + (t-2)] + y_{t-3} + \varepsilon_{t-2} + \varepsilon_{t-1} + \varepsilon_t \\ &= \ldots \\ &= y_0 + \alpha t + \beta \frac{t(t+1)}{2} + \sum_{i=1}^{t} \varepsilon_i, \end{aligned}$$

   which contains a quadratic trend. Therefore, the joint null $H_0 : \delta = \beta = 0$ is more appropriate when testing for a unit root in a time series with a linear trend since it allows the model to share the same trend features whether $\delta = 0$ or not.

2. (a) The model can be written as $\phi(L) y_t = \varepsilon_t$, where

$$\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p.$$

   If $\phi(z) = 1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p$ has a unit root at 1, then $\phi(1) = 1 - \phi_1 - \phi_2 - \ldots - \phi_p = 0$, or $\sum_{i=1}^{p} \phi_i = 1$.

   (b) Subtracting y_{t-1} from both sides of the equation gives

$$\Delta y_t = (\phi_1 - 1) y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + \ldots + \phi_p y_{t-p} + \varepsilon_t.$$

   Adding $\phi_2 y_{t-1}$ to the first term and subtracting $\phi_2 y_{t-1}$ from the second term yields

$$\Delta y_t = (\phi_1 + \phi_2 - 1) y_{t-1} - \phi_2 (y_{t-1} - y_{t-2}) + \phi_3 y_{t-3} + \ldots + \phi_p y_{t-p} + \varepsilon_t.$$

   Then adding $\phi_3 y_{t-1}$ to the first term and subtracting $\phi_3 y_{t-1}$ from the second term gives

$$\Delta y_t = (\phi_1 + \phi_2 + \phi_3 - 1) y_{t-1} - (\phi_2 + \phi_3)(y_{t-1} - y_{t-2}) - \phi_3 (y_{t-2} - y_{t-3}) + \phi_4 y_{t-4} + \ldots + \phi_p y_{t-p} + \varepsilon_t.$$

   Continuing in this fashion, we can obtain the desired representation. This exercise shows that if $\{y_t\}$ has a unit root, then $\{\Delta y_t\}$ is an $AR(p-1)$ process, and testing the hypothesis of a unit root is equivalent to testing $\phi_1^* = \sum_{i=1}^{p} \phi_i - 1 = 0$.

3. (a) $ARIMA(0, 1, 1)$

(b) Note that y_t is a unit-root process, and we cannot invert the AR polynomial.

$$\begin{aligned} y_t - y_{t-1} &= -1.2 + (1 - 0.6L)\,\varepsilon_t, \\ y_{t-1} - y_{t-2} &= -1.2 + L\,(1 - 0.6L)\,\varepsilon_t, \\ &\vdots \\ y_2 - y_1 &= -1.2 + L^{t-2}\,(1 - 0.6L)\,\varepsilon_t, \\ y_1 - y_0 &= -1.2 + L^{t-1}\,(1 - 0.6L)\,\varepsilon_t. \end{aligned}$$

Summing the above equations gives

$$y_t - y_0 = -1.2t + \left(1 + L + \ldots + L^{t-1}\right)(1 - 0.6L)\,\varepsilon_t.$$

It follows that

$$E\,(y_t) = y_0 - 1.2t = 1 - 1.2t,$$

which diverges as t increases.

(c) x_t is stationary. To see why $E\,(x_t)$ converges for t large, you can also use the same argument as in (b) to obtain

$$x_t = x_0 - \left(1 + 0.95 + 0.95^2 + \ldots + 0.95^{t-1}\right) \times 1.2 + \left(1 + 0.95L + \ldots + 0.95^{t-1}L^{t-1}\right)(1 - 0.6L)\,\varepsilon_t.$$

Alternatively, we can rewrite the model as

$$\begin{aligned} x_t &= -\,(1 - 0.95L)^{-1} \times 1.2 + (1 - 0.95L)^{-1}\,(1 - 0.6L)\,\varepsilon_t \\ &= -\left(1 + 0.95 + 0.95^2 + \ldots\right) \times 1.2 + \left(1 + 0.95L + 0.95^2L^2 + \ldots\right)(1 - 0.6L)\,\varepsilon_t, \end{aligned}$$

if t is sufficiently large, and it follows that

$$E\,(x_t) = -\frac{1.2}{1 - 0.95} = -24.$$

The mean of x_t converges since $\sum_{i=0}^{\infty} 0.95^i$ is a convergent series.
Remark: You can also show the results in (b) and (c) by recursive substitution.

4. (a) y_t is stationary since it only contains a finite-order moving average component. y_t is also invertible, since the roots of $\theta\,(z) = 1 + 0.5z = 0$ and $\Theta\,(z) = 1 + 0.3z = 0$ both lie outside of the unit circle.
Remark: For the seasonal polynomial, you only need to solve $\Theta\,(z) = 1 + 0.3z = 0$ (i.e., treating L^4 as z), which gives $z = -\frac{10}{3}$, and $\left|-\frac{10}{3}\right| > 1$. You will reach the same conclusion about invertibility if you solve $1 + 0.3z^4 = 0$, which is more complicated since there are four

roots. The following results are not required, but in case you are curious,

$$z = \left(-\frac{10}{3}\right)^{\frac{1}{4}} = \left(\frac{10}{3}\right)^{\frac{1}{4}} (-1)^{\frac{1}{4}}.$$

By De Moivre's Theorem, it can be shown that the four roots of $x^4 + 1 = 0$ are

$$x = \pm\frac{1+i}{\sqrt{2}}, \pm\frac{1-i}{\sqrt{2}},$$

and therefore

$$z = \pm\left(\frac{10}{3}\right)^{\frac{1}{4}}\left(\frac{1+i}{\sqrt{2}}\right), \pm\left(\frac{10}{3}\right)^{\frac{1}{4}}\left(\frac{1-i}{\sqrt{2}}\right),$$

or $z = \pm 0.955 \pm 0.955i$, Since x has a unit modulus, $\frac{10}{3} > 1$, all four roots are outside the unit circle. Now you see that if $|\Theta| < 1$, then $\Theta\left(L^4\right) = 1 + \Theta L^4$ will be invertible.

(b) Note that

$$y_t = \left(1 + 0.5L + 0.3L^4 + 0.15L^5\right)\varepsilon_t = \varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-4} + 0.15\varepsilon_{t-5}.$$

Then we obtain $E(y_t) = 0$, and

$$Var(y_t) = E\left(\varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-4} + 0.15\varepsilon_{t-5}\right)^2 = \left(1 + 0.5^2 + 0.3^2 + 0.15^2\right)\sigma^2 = 1.3625\sigma^2,$$

$$\begin{aligned}\gamma_1 &= E(y_t y_{t-1}) = E\left[\left(\varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-4} + 0.15\varepsilon_{t-5}\right)\left(\varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.3\varepsilon_{t-5} + 0.15\varepsilon_{t-6}\right)\right] \\ &= 0.5E\left(\varepsilon_{t-1}^2\right) + 0.15 \times 0.3 \times E\left(\varepsilon_{t-5}^2\right) = 0.545\sigma^2,\end{aligned}$$

$$\begin{aligned}\gamma_3 &= E(y_t y_{t-3}) = E\left[\left(\varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-4} + 0.15\varepsilon_{t-5}\right)\left(\varepsilon_{t-3} + 0.5\varepsilon_{t-4} + 0.3\varepsilon_{t-7} + 0.15\varepsilon_{t-8}\right)\right] \\ &= 0.3 \times 0.5E\left(\varepsilon_{t-4}^2\right) = 0.15\sigma^2,\end{aligned}$$

$$\begin{aligned}\gamma_4 &= E(y_t y_{t-4}) = E\left[\left(\varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-4} + 0.15\varepsilon_{t-5}\right)\left(\varepsilon_{t-4} + 0.5\varepsilon_{t-5} + 0.3\varepsilon_{t-8} + 0.15\varepsilon_{t-9}\right)\right] \\ &= 0.3E\left(\varepsilon_{t-4}^2\right) + 0.15 \times 0.5E\left(\varepsilon_{t-5}^2\right) = 0.375\sigma^2,\end{aligned}$$

$$\begin{aligned}\gamma_5 &= E(y_t y_{t-5}) = E\left[\left(\varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-4} + 0.15\varepsilon_{t-5}\right)\left(\varepsilon_{t-5} + 0.5\varepsilon_{t-6} + 0.3\varepsilon_{t-9} + 0.15\varepsilon_{t-10}\right)\right] \\ &= 0.15E\left(\varepsilon_{t-5}^2\right) = 0.15\sigma^2.\end{aligned}$$

Therefore, the ACFs are given by

$$\begin{aligned}
&\rho_0 = 1, \\
&\rho_1 = 0.545/1.3625 = 0.4, \\
&\rho_3 = 0.15/1.3625 = 0.11 \\
&\rho_4 = 0.375/1.3625 = 0.275 \\
&\rho_5 = 0.15/1.3625 = 0.11 \\
&\rho_h = 0, \text{ for } h \neq 0, 1, 3, 4, 5.
\end{aligned}$$

(c) For the pure MA(1) model, $y_t = (1 + 0.5L)\,\varepsilon_t$, it is immediate that $\rho_0 = 1$, $\rho_1 = 0.5/\left(1 + 0.5^2\right) = 0.4$, and $\rho_h = 0$, for all $h \geq 2$. For the pure SMA(1) model, $y_t = \left(1 + 0.3L^4\right)\varepsilon_t$, we have $\rho_0 = 1$, $\rho_4 = 0.3/\left(1 + 0.3^2\right) = 0.275$, and $\rho_h = 0$ otherwise. In comparison, the model that contains both MA(1) and SMA(1) components have non-zero ACFs at lags 1, 3, 4, and 5. The non-zero ACF at lag 1 is the same as that of the pure MA(1) model, at lag 4 is the same as that of the pure SMA(1) model, at lags 3 and 5 is the same as the product of the non-zero ACFs of the MA(1) and SMA(1) models and can be viewed as the interaction effect of the lag-1 and lag-4 serial dependence.

Question 5

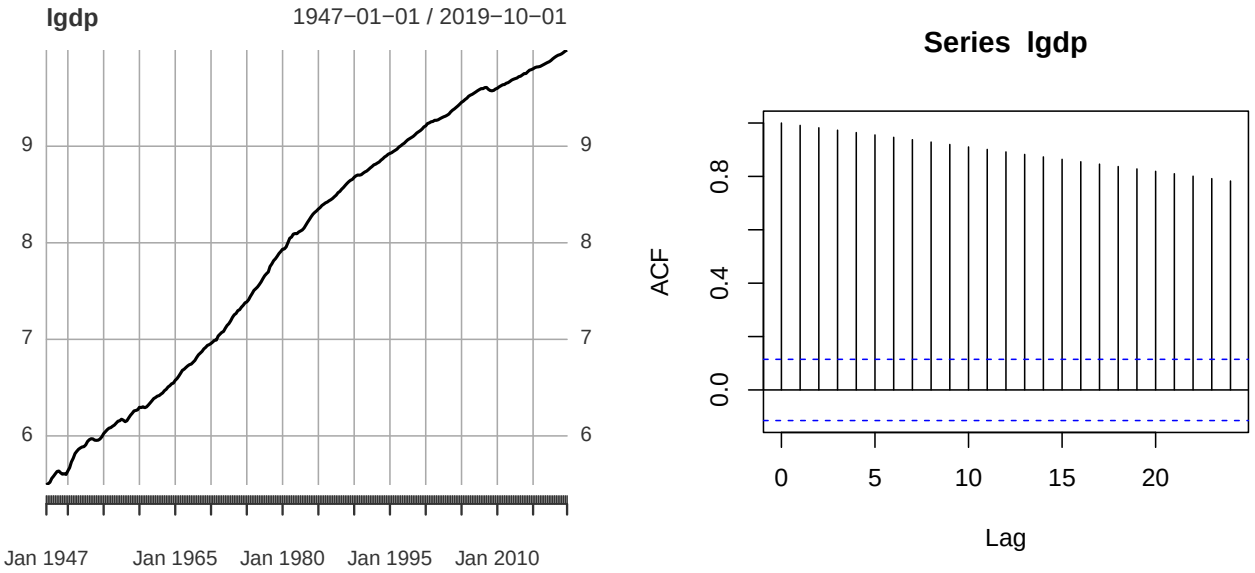
(a) The log GDP series is clearly trended, but we need to determine whether it is better modeled as a deterministic trend or stochastic trend. To this end, we use the ADF regression with an intercept and a trend and consider the phi3 statistic for the joint hypotheses. Since the phi3 statistic is less than the critical values at all conventional significance levels, we fail to reject the null of a unit root and zero trend coefficient. Our conclusion is that *lgdp* is likely to be difference stationary provided that the difference of lgdp is I(0), which we will examine next in part (b).

Remark: In practice, it is advisable to check the results using different numbers of lags based on AIC/BIC, significance of the lags (sequential t tests), and residuals of the ADF regressions. These steps were omitted in this solution guide. Inspecting the ACF/PACF of the differenced series can also aid in lag selection. For a quarterly series with seasonality, the maximum lag should be at least 4. In this note, I used information criteria with up to 6 lags for simplicity, but you can explore other values and examine the residuals and significance of the lags. A rule-of-thumb value for p_{max} suggested by Schwert (1989) is $[12(T/100)^{1/4}]$, where $[a]$ denotes the integer part of a. At any rate, keep in mind that if p is too small, the remaining serial correlation in the errors will bias the test; if p is too large, the test will lack statistical power.

```
library(quantmod)
library(urca)
library(forecast)
getSymbols("GDP", src="FRED")
```

```
## [1] "GDP"
```

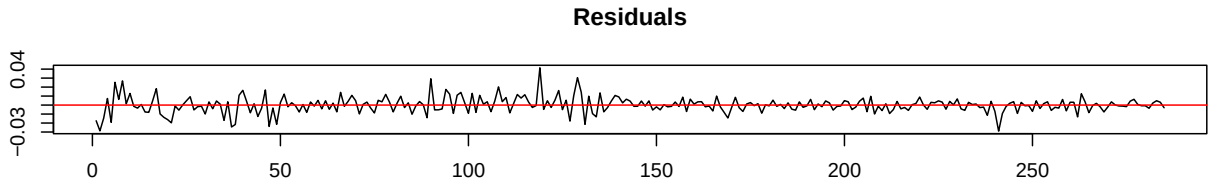
```
lgdp<-log(GDP)['/2019']
par(mfrow=c(1,2))
plot(lgdp)
acf(lgdp)
```

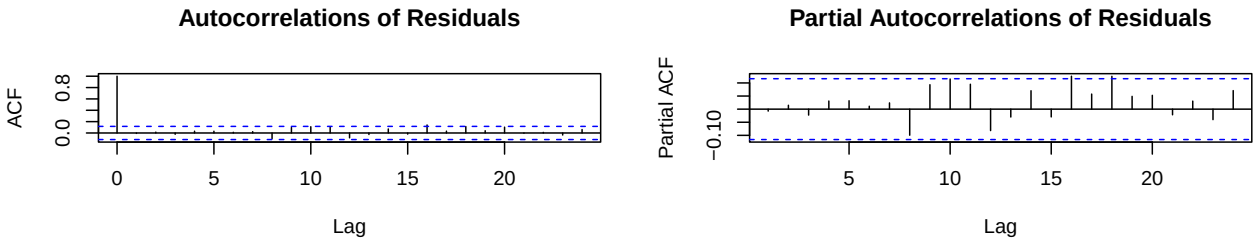


```r
lgdp_urdf<-ur.df(lgdp, type="trend", lags=6, selectlags="AIC")
summary(lgdp_urdf)
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.029048 -0.004848  0.000116  0.004265  0.041447
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.573e-04  1.761e-02  -0.054  0.95669
## z.lag.1      2.080e-03  3.274e-03   0.635  0.52582
## tt          -5.052e-05  5.579e-05  -0.906  0.36596
## z.diff.lag1  4.222e-01  6.004e-02   7.031 1.61e-11 ***
## z.diff.lag2  2.060e-01  6.446e-02   3.195  0.00156 **
## z.diff.lag3 -1.031e-01  6.562e-02  -1.571  0.11736
## z.diff.lag4 -2.115e-02  6.486e-02  -0.326  0.74463
## z.diff.lag5 -1.541e-01  6.372e-02  -2.418  0.01624 *
## z.diff.lag6  1.158e-01  5.961e-02   1.942  0.05312 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.009076 on 276 degrees of freedom
## Multiple R-squared:  0.3172, Adjusted R-squared:  0.2974
## F-statistic: 16.03 on 8 and 276 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic is: 0.6352 10.2351 2.5761
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

```r
plot(lgdp_urdf)
```



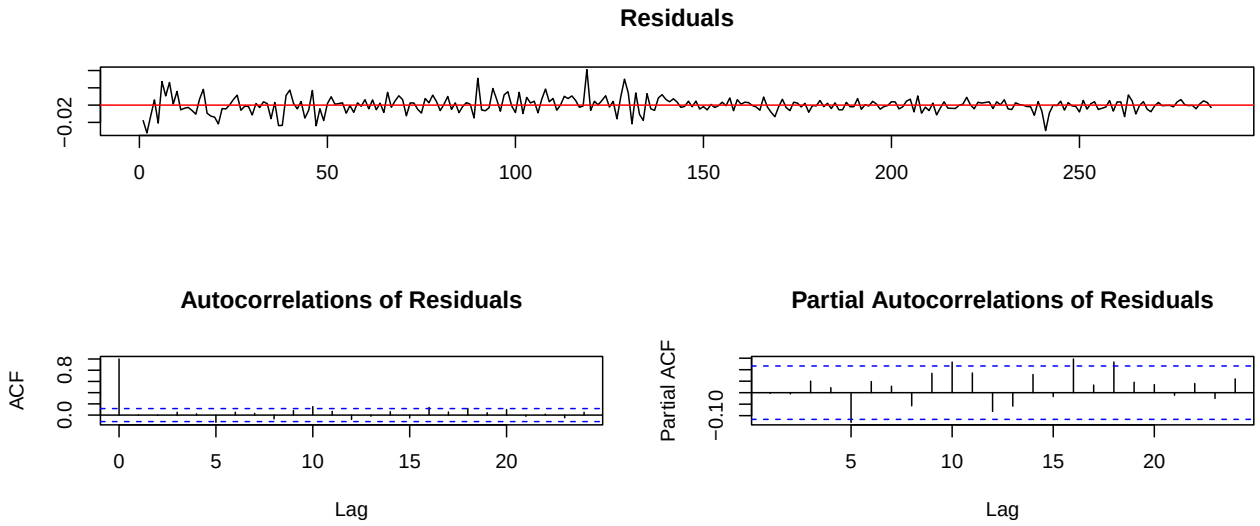



```
lgdp_urdf<-ur.df(lgdp, type="trend", lags=6, selectlags="BIC")
summary(lgdp_urdf)
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.032217 -0.004679 -0.000215  0.004328  0.041069
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.522e-04  1.748e-02  -0.009 0.993058
## z.lag.1      1.893e-03  3.205e-03   0.591 0.555249
## tt          -4.698e-05  5.419e-05  -0.867 0.386742
## z.diff.lag1  4.206e-01  5.926e-02   7.097 1.06e-11 ***
## z.diff.lag2  2.124e-01  6.322e-02   3.359 0.000891 ***
## z.diff.lag3 -1.573e-01  5.969e-02  -2.635 0.008878 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.009151 on 279 degrees of freedom
## Multiple R-squared:  0.2983, Adjusted R-squared:  0.2858
## F-statistic: 23.73 on 5 and 279 DF,  p-value: < 2.2e-16
##
```

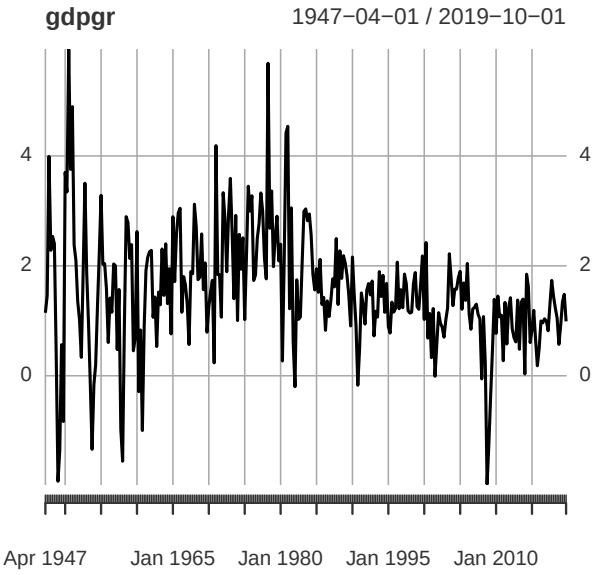
```
##
## Value of test-statistic is: 0.5906 15.8734 2.6198
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

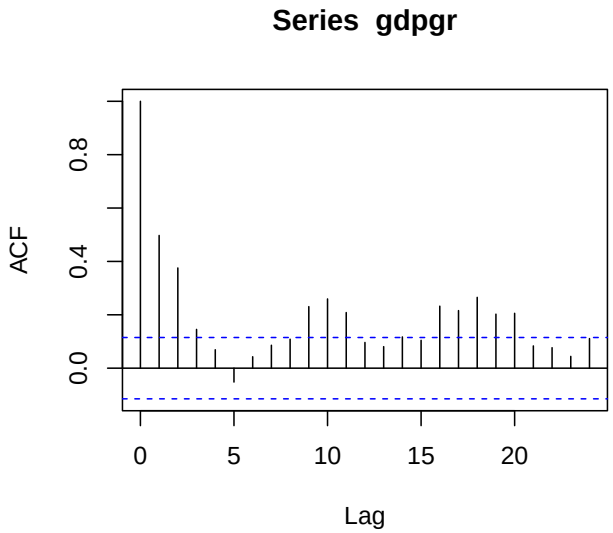
```
plot(lgdp_urdf)
```



(b) Both the time series and ADF plots suggest that GDP growth may be stationary (although the variance does not seem constant). The phi1 statistic is greater than the critical value even at the 1% level. We reject the null of a unit root (with a zero intercept) in favor of the alternative that GDP growth is I(0). The above results provide evidence that the *lgdp* series is difference stationary, and hence it is appropriate to apply stationary ARMA models to the difference of *lgdp*.

```
gdpgr<-100*diff(lgdp)[-1]
par(mfrow=c(1,2))
plot(gdpgr)
acf(gdpgr)
```






```
gdpgr_urdf<-ur.df(gdpgr, type="drift", lags=6, selectlags="AIC")
summary(gdpgr_urdf)
```

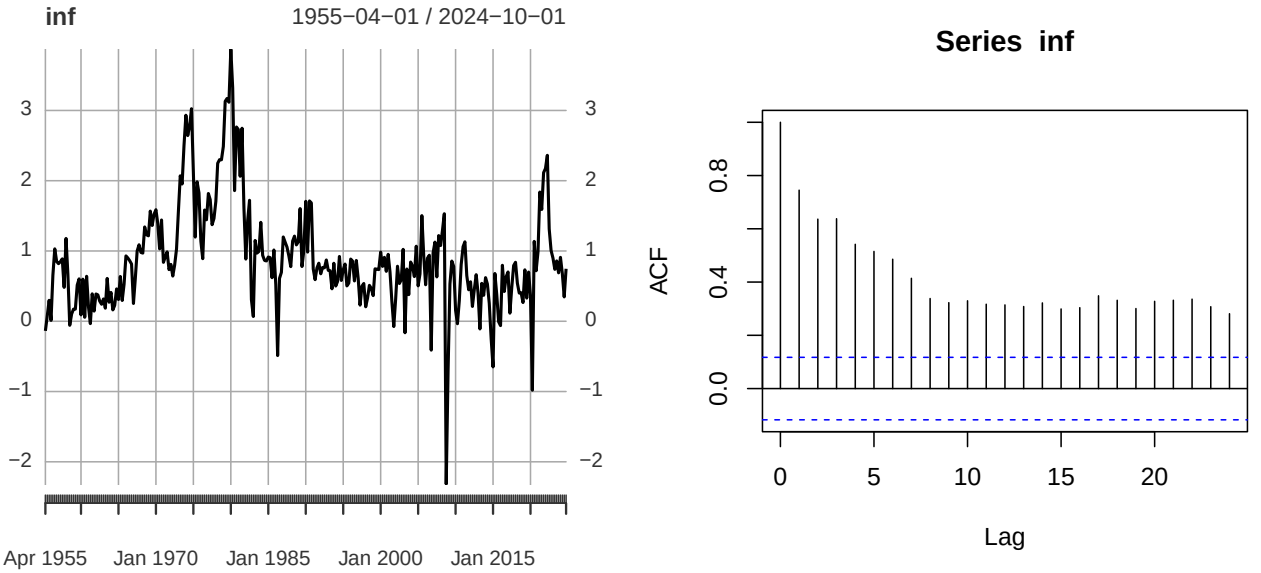
```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.9973 -0.5186  0.0192  0.4085  4.1188
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.60130    0.14004   4.294 2.44e-05 ***
## z.lag.1     -0.39158    0.08361  -4.684 4.43e-06 ***
## z.diff.lag1 -0.17450    0.08587  -2.032  0.04310 *
## z.diff.lag2  0.05801    0.07875   0.737  0.46194
## z.diff.lag3 -0.03902    0.07457  -0.523  0.60118
## z.diff.lag4 -0.02366    0.07007  -0.338  0.73585
## z.diff.lag5 -0.19320    0.06688  -2.889  0.00418 **
## z.diff.lag6 -0.10393    0.05881  -1.767  0.07830 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.9059 on 276 degrees of freedom
## Multiple R-squared:  0.3124, Adjusted R-squared:  0.295
## F-statistic: 17.92 on 7 and 276 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic is: -4.6835 10.971
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

(c) The inflation series does not exhibit a clear upward or downward trend, but it is highly persistent according to the ACF plot. Accordingly, we use the ADF regression with a drift. With the lag order selected by AIC, the phi1 statistic suggests that we fail to reject the null of a unit root at the 5% level, though it is rejected at the 10% level. The KPSS test statistic, computed with five lags using the "short" option, exceeds the critical value even at the 1% level. Thus, the KPSS test leads us to reject the null of stationarity.

Turning to the first difference of inf, both the ADF and KPSS tests suggest that it is stationary. Overall, we find evidence that the US inflation series is I(1), aligning with findings in the literature.

```
cpi<-getSymbols("CPALTT01USQ661S",src="FRED", auto.assign=FALSE)
inf<-100*diff(log(cpi))[-1]
par(mfrow=c(1,2))
plot(inf)
acf(inf)
```




```
inf_urdf<-ur.df(inf, type="drift", lags=6, selectlags="AIC")
summary(inf_urdf)
```

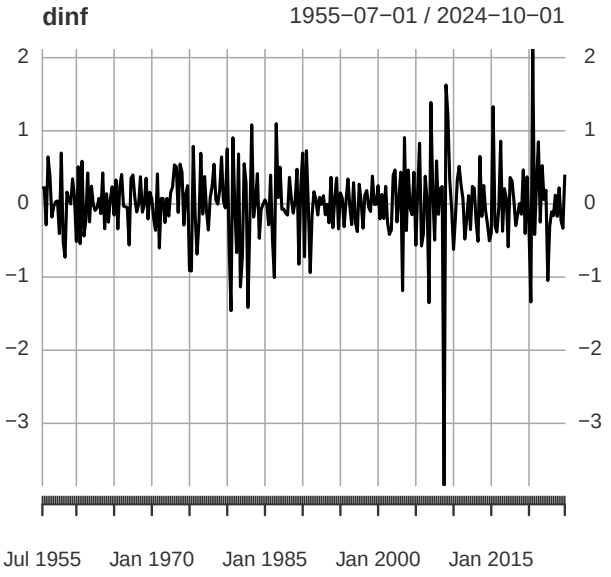
```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.5603 -0.2226  0.0260  0.2721  1.3280
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.12574    0.04921   2.555   0.0112 *
## z.lag.1     -0.14108    0.04475  -3.153   0.0018 **
## z.diff.lag1 -0.28168    0.06735  -4.182 3.92e-05 ***
## z.diff.lag2 -0.28971    0.06889  -4.206 3.56e-05 ***
## z.diff.lag3  0.01092    0.06585   0.166   0.8684
## z.diff.lag4 -0.11178    0.06070  -1.842   0.0667 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4685 on 266 degrees of freedom
## Multiple R-squared:  0.2318, Adjusted R-squared:  0.2174
## F-statistic: 16.06 on 5 and 266 DF,  p-value: 7.883e-14
##
##
## Value of test-statistic is: -3.1525 4.9697
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

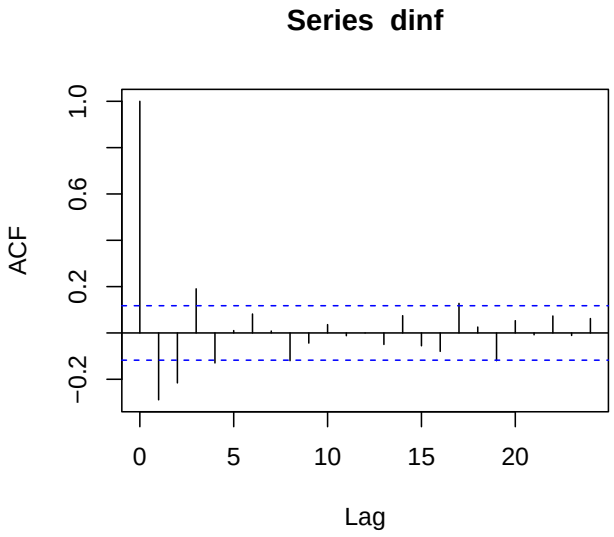
```
inf_kpss<-ur.kpss(inf, type="mu", lags="short")
summary(inf_kpss)
```

```
##
## ########################
## # KPSS Unit Root Test #
## ########################
##
## Test is of type: mu with 5 lags.
##
```

```
## Value of test-statistic is: 0.6238
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```
dinf<-diff(inf)[-1]
par(mfrow=c(1,2))
plot(dinf)
acf(dinf)
```






```
dinf_urdf<-ur.df(dinf, type="drift", lags=6, selectlags="AIC")
summary(dinf_urdf)
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.5914 -0.2115  0.0437  0.2504  1.4355
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.001034   0.028987  -0.036   0.9716
## z.lag.1     -1.953471   0.172317 -11.337  < 2e-16 ***
```

```
## z.diff.lag1  0.572179   0.144678   3.955 9.83e-05 ***
## z.diff.lag2  0.198601   0.103039   1.927   0.0550 .
## z.diff.lag3  0.152951   0.060381   2.533   0.0119 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4772 on 266 degrees of freedom
## Multiple R-squared:  0.6912, Adjusted R-squared:  0.6865
## F-statistic: 148.8 on 4 and 266 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic is: -11.3365 64.2591
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```
dinf_kpss<-ur.kpss(dinf, type="mu", lags="short")
summary(dinf_kpss)
```

```
##
## ########################
## # KPSS Unit Root Test #
## ########################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 0.0388
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```